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Osteology of the pelvic limb

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Course outline

Introduction

1. hip bone
2. Femur
3. Patella
4. Leg skeleton:
1. Tibia
2. Fibula
5. Foot skeleton

Introduction

The skeleton of the pelvic limb(lower limb) includes the following paired bones:

- **The femur** and the patella (kneecap) form the skeleton of the thigh.
- **The tibia** and the fibula (peroneus) form the skeleton of the leg.
- **The skeleton of the foot**.

The skeleton of the pelvic limb is connected to the trunk by the pelvic girdle, which is made up of the two iliac or hip bones, the sacrum and coccyx.

The pelvic limb presents three longitudinal skeletal axes who are:

- *The anatomical axis of the femoral diaphysis*: it is oblique downwards and inwards, and makes an angle of 9° with the vertical.
- *The mechanical axis*(or functional) of the thigh: it runs from the center of the femoral head to the center of the knee (top of the patella). It makes an angle of 3° with the vertical.
- *The anatomical axis of the leg*: it runs from the center of the knee to the middle of the ankle. It makes an angle of 3° with the vertical.

Thus, the angle formed by th axis of the femoral diaphysis and the axis of the leg is open outwards, giving the physiological genu-valgum (= 175°): a more closed angle gives the pathological genu-valgum (thighs and legs in X), and a more open angle gives the pathological genu-varum (thighs and legs in O).

1. hip bone

The hip bone (iliac bone) is a flat bone, shaped like a two-bladed helix, it connects the spine and the pelvic limb.

Embryologically, the hip bone is made up of the junction of 3 bone pieces:

- Ilium
- Ischium
- Pubis

The three parts meet in the acetabulum, forming a "Y" shape.

1.1. Set up

- **Outside**: the joint cavity of the bone.
- **Down :**the foramen of the bone.
- **Backwards:**the notched edge of the bone.

1.2. Descriptive Anatomy

The hip bone has to be described: two faces, four edges and four angles.

- **The side**

  The side face presents, from top to bottom:

  - **The gluteal surface**(The lateral iliac fossa) presents two gluteal lines: anterior and posterior, which delimit the insertion fields of the gluteal muscles: gluteus minimus, gluteus medius and gluteus maximus.
  - **The acetabulum**(The acetabular cavity) is a deep and spheroid excavation, limited by a prominent rim, it is the acetabular limbus (the acetabular brow),

which has three notches (the lower notch is wide and deep) corresponding to the points of junction of the three bony pieces constituting the hip bone: ilium, pubis and ischium, which meet by drawing a "Y".

The acetabulum has two distinct parts:

- *The central part*, non-articular, it is the acetabular fossa (background)
- *The peripheral part*, articular surface covered with hyaline-type cartilage, crescent-shaped, it is the semi-lunar surface.
    - **The obturator foramen (**The ischiopubic foramen (also called the obturator foramen) is a large opening located below the acetabulum, closed by two obturator membranes, medial and lateral. It is oval and narrow in men, and triangular and wide in women.

This foramen is limited by the obturator frame consisting of:

- At the top: the cranial (or horizontal) branch of the pubis which presents the obturator groove (or subpubic groove).
- Inside: the body of the pubis (or quadrilateral plate).
- Behind: the ischial tuberosity.
- Below: the ischiopubic ramus.

- **Medial face** is flared and supports the abdominal viscera below.

It is crossed by a curved ridge, the arched line (unnamed or ridge of the upper strait) which divides it into two parts:

- At the top, the iliac fossa, which provides insertion for the iliacus muscle.
- Below, the obturator foramen which presents along its upper border the pecten pubis (pectineal crest) and at the level of the cranial branch of the pubis the obturator groove.

Behind the arcuate line, a protruding surface appears, articulating with the sacrum; this is the auricular surface, behind which the sacroiliac ligaments attach.

- **Edges**
    - **The upper edge** is the iliac crest, palpable under the skin and easily identifiable; it is a preferred site for bone punctures and harvesting bone grafts.
    - **The bottom edge** articulates with the hip bone on the opposite side by its anterior segment forming the pubic symphysis.
    - **The anterior edge** from top to bottom presents: the anterior superior iliac spine, the innominate notch, the anterior inferior iliac spine, the iliopectineal eminence and the angle of the pubis.
    - **The rear edge** presents from top to bottom: the postero-superior iliac spine, the small innominate notch, the postero-inferior iliac spine, the greater ischial notch, the ischial spine, the lesser ischial notch and the ischial tuberosity.

- **Angles**

The anterosuperior angle: corresponds to the anterior superior iliac spine.
The posterosuperior angle: corresponds to the posterior superior iliac spine.
The anteroinferior angle: corresponds to the pubic spine.
Posterior-inferior: corresponds to the ischial tuberosity.

Note: the bony pelvis
The two hip bones, together with the sacrum and coccyx, form the bony pelvis.
The pelvis is divided by the plane of the superior strait into the greater pelvis at the top and the lesser pelvis (or obstetrical pelvis or pelvic excavation) at the bottom.
*The plan of the upper strait*It is formed by the union of two laterally arched lines, the promontory of the sacrum posteriorly and the superior border of the pubic symphysis anteriorly. It presents the superior opening of the pelvis, the main area to be cooled during childbirth.
Knowledge of the limits and diameters of the pelvis (pelvimetry) is important in obstetrics.

2. Femur

The femur is a long bone that forms the skeleton of the thigh. It articulates at the top with the hip bone, and at the bottom with the tibia and patella.

2.1. Set up :

- **Up :**the angled end.
- **Inside:**the articular surface of this extremity.
- **Backwards:**the most prominent edge.

2.2. Descriptive Anatomy

The femur is presented to be described*a diaphysis*and two epiphyses.

- **Shaft**

The diaphysis presents: three surfaces (anterior, posterolateral and posteromedial) which give insertion to the quadriceps muscle, and three borders (lateral, medial and posterior).
The rear edge, also called the rough line, is a prominent and rugged ridge.
The linea aspera trifurcates at the top, and at the bottom it bifurcates, delimiting a triangular space, this is the popliteal surface.

- **Proximal epiphysis**

The proximal epiphysis includes: the femoral head, the femoral neck, the greater trochanter and the lesser trochanter.

- **Femoral head**

The femoral head is a rounded protrusion representing 2/3 of a sphere, entirely covered with cartilage except at the fovea capitis (or fovea of the round ligament) where the ligament of the femoral head (or round ligament) attaches.
The head articulates with the acetabulum of the hip bone.

- **Neck of the femur :**

The neck of the femur makes an angle of inclination of 125° to 130° with the axis of the diaphysis.
The more open angle : coxa valga.
The more closed angle : coxa vara

- **The greater trochanter**

The greater trochanter is a lateral quadrangular projection, its medial face presents the trochanteric fossa (digital fossa).

- **The lesser trochanter**is a medial and conical projection.

The two major and minor tubercles are connected anteriorly by the intertrochanteric line (or anterior intertrochanteric line) and posteriorly by the intertrochanteric crest (or posterior intertrochanteric line).

- **Distal epiphysis**

The distal epiphysis presents to describe the patellar surface of the femur (the femoral trochlea) and the femoral condyles.

- **Femoral condyles**

The femoral condyles are two articular eminences, each curved and curled upon itself. The condyles correspond to the tibial plateau below and the patella anteriorly.
The two condyles, medial and lateral, are separated from each other posteriorly by the intercondylar fossa where the cruciate ligaments of the knee attach.
The lateral face of each condyle is occupied by a tuberosity where the collateral ligaments attach: the lateral epicondyle at the level of the lateral condyle and the medial epicondyle at the level of the medial condyle.

- **femoral trochlea**

The trochlea is located on the anterior face of the distal femoral epiphysis, it articulates with the posterior face of the patella.

3. Patella

The patella (kneecap) is a sesamoid bone, flattened and triangular, located in the tendon of the quadriceps femoris muscle, on the anterior part of the knee.
She presents the following to describe:
Two sides *previous* which is convex and posterior, articular with the femoral trochlea.
Two edges lateral and medial.
A base superior where the quadriceps tendon attaches.
A summit inferior, where the patellar ligament (or patellar tendon) attaches.

4. Leg skeleton

The skeleton of the leg is made up of two long bones, the tibia and the fibula (or peroneus).

4.1. TIBIA

The tibia is the medial and main bone of the leg, articular at the top with the femur and fibula, and at the bottom with the talus (astragalus).

4.1.1. Set up

- **Down :** the less voluminous end.

- **Inside:**the apophysis of this extremity.
- **Ahead :** the sharpest edge of the bone.

4.1.2. Descriptive Anatomy

The femur has to be described as having a diaphysis and two epiphyses.

- ## Shaft

The diaphysis is prismatic and triangular, with the following to be described: 3 faces and 3 edges.

The side provides insertion for the tibialis anterior (or tibial) muscle.

The medial surface is smooth and subcutaneous, mostly lacking muscular insertion.

The posterior side is crossed obliquely by the line of the soleus muscle (oblique line).

The edges : There are three of them: medial, lateral (interosseous) and anterior. The latter is subcutaneous, and also called the tibial crest.

- ## Proximal epiphysis

The proximal epiphysis is large, shaped like a quadrangular pyramid with its base at the top.

- **Base (tibial plateau)**

The tibial plateau presents:

- Two superior tibial articular surfaces (medial and lateral glenoid cavities) articulate with the femoral condyles.
- *Intercondylar area* (inter-glenoid space), occupied by two intercondylar tubercles (or tibial spines), medial and lateral.

The anterior face presents the anterior tibial tuberosity which gives insertion to the patellar ligament.

The posterior side : externally, it presents the fibular articular surface.

The side faces formed by two tibial tuberosities, lateral and medial.

- ## Distal epiphysis

The inferior surface of the distal epiphysis presentstwo continuous articular surfaces (tibial plafond) articular joint with the talus

- **The lower tibial articular surface**
- **The articular surface of the medial malleolus**

The fibular notch which is a notch occupying the lateral face of this epiphysis, it applies to the distal end of the fibula.

The medial malleolus which is a process which continues down the medial face of the bone, and articulates with the talus.

4.2. FIBULA

The fibula is the lateral and slender bone of the leg; it articulates at the top with the tibia, and at the bottom with the tibia and talus. It is formed of a diaphysis and two epiphyses.

4.2.1. Set up :

- **Down :** the flattened end.
- **Inside:**the articular facet of this extremity.

- **Backwards:**the rough dimple at this end.

4.2.2. Descriptive Anatomy

- **Shaft**

The diaphysis is triangular, and has : three faces (lateral, medial and posterior) and three borders (lateral, medial and anterior).

- **Proximal epiphysis**

The proximal epiphysis is also called the head of the fibula ; it articulates with the tibia.

- **Distal epiphysis:**

The distal epiphysis forms the lateral malleolus, which is longer and larger than the medial malleolus and extends lower. Its medial surface articulates with the talus.

5. FOOT BONES

The foot is made up of 26 bones, which are divided into three groups:

- *The tarsus*consisting of seven bones
- *The metatarsus*consisting of five bones
- *The phalanges of the toes*, are 14 in number.

5.1. tarsus

The tarsus is arranged in two rows connected by the Chopart mid-tarsal joint.

The first row (posterior tarsus) includes: talus (astragalus) and calcaneus (calcaneus).

The second row (anterior tarsus) comprises five juxtaposed bones which are from inside out: navicular bone (scaphoid), 1st cuneiform (or medial cuneiform), 2nd cuneiform (or intermediate cuneiform), 3rd cuneiform (or lateral cuneiform) and cuboid bone.

5.2. metatarsus

The metatarsus is formed of five long bones, the metatarsals, listed from the inside out.

The metatarsus is connected to the tarsus by the Lisfranc tarsometatarsal joint.

5.3. phalanges

There are 14 phalanges, three for each toe (proximal, middle and distal phalanges), except for the hallux (or big toe) which has two phalanges.

Note :

*The sesamoid bones of the foot*are always located on the plantar surface of the foot, two of them are constant, located opposite the head of the 1st metatarsal.

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